CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2023 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **How is an Object created in Java?**

Object Creation: Uses the `new` operator:

```
ClassName obj = new ClassName();
```

Steps: (1) *Declaration* (associates variable name with object type), (2) *Instantiation* (`new` allocates memory on the Heap), (3) *Initialization* (invokes the constructor).

Q2 (3M) **Compare JDK, JRE, and JVM.**

Relationship:

- **JVM (Java Virtual Machine):** The runtime engine that interprets and executes bytecode.
- **JRE (Java Runtime Environment):** Contains JVM + Java Standard Class Libraries; sufficient to run Java programs.
- **JDK (Java Development Kit):** Contains JRE + Development Tools (`javac`, `javadoc`, `jar`, `debugger`); needed to write and compile programs.

Q3 (3M) **Difference between JDBC and ODBC.**

Differences: ODBC is a C/C++ platform-dependent API designed originally for Windows. JDBC is a platform-independent pure Java API tailored for Java applications and cross-platform RDBMS connectivity.

Q4 (3M) **Write differences between JSP and Servlet.**

Differences: Servlets are Java code with embedded HTML strings (ideal for business logic and controllers). JSP is HTML markup with embedded Java scriptlets (ideal for presentation and views in MVC).

Q5 (3M) **What is Method Overloading vs Method Overriding?**

Comparison: Overloading occurs in the same class with identical method names but different parameter lists (compile-time polymorphism). Overriding occurs in subclasses with identical method signatures and return types (runtime dynamic dispatch).

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **What is an Applet? Describe the complete Life Cycle of an Applet.**

Applet Life Cycle: `init()` → `start()` → `paint(Graphics g)` → `stop()` → `destroy()`.

Q7 (7.5M) **Explain Interface in Java. How is multiple inheritance achieved using interfaces?**

Explanation: A class can implement multiple interfaces simultaneously:

```
interface Printable { void print(); }
interface Showable { void show(); }
class Document implements Printable, Showable {
    public void print() { System.out.println("Printing..."); }
    public void show() { System.out.println("Showing..."); }
}
```

Q8 (7.5M) **Explain Exception Handling in Java with try-catch-finally block.**

Handling: Code capable of raising runtime exceptions is placed inside `try`, recovered in `catch`, and resource deallocation performed in `finally`.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **What are Packages in Java? How is a user-defined package created and imported? Explain with complete code example.**

Creating & Importing User-Defined Package:

```
// File: mypack/Calculator.java
package mypack;

public class Calculator {
    public int add(int a, int b) { return a + b; }
    public int multiply(int a, int b) { return a * b; }
}

// File: TestPackage.java
import mypack.Calculator;

public class TestPackage {
    public static void main(String[] args) {
        Calculator calc = new Calculator();
        System.out.println("Sum: " + calc.add(10, 20));
        System.out.println("Product: " + calc.multiply(10, 20));
    }
}
```

Q10 (15M) **Define Thread. How to create a thread in Java? Write a program that executes two threads simultaneously.**

Complete Dual Thread Program:

```
class ThreadOne extends Thread {
    public void run() {
        for(int i = 1; i ⩽ 5; i++) {
            System.out.println("Thread 1 executing count: " + i);
            try { Thread.sleep(400); } catch(Exception e) {}
        }
    }
}

class ThreadTwo extends Thread {
    public void run() {
        for(int j = 10; j ⩽ 15; j++) {
            System.out.println("Thread 2 executing count: " + j);
            try { Thread.sleep(400); } catch(Exception e) {}
        }
    }
}

public class DualThreadDemo {
    public static void main(String[] args) {
        ThreadOne t1 = new ThreadOne();
        ThreadTwo t2 = new ThreadTwo();
        t1.start();
        t2.start();
    }
}
```

Q11 (15M) **Explain JDBC Architecture and steps involved in connecting a Java application to a Database. Write complete program.**

Steps: (1) Load Driver, (2) Create Connection, (3) Create Statement, (4) Execute Query, (5) Process ResultSet, (6) Close Connection.

Q12 (15M) **Explain HTML form controls and write code for an Admission Form.**

Admission Form: Includes text fields, dropdown selects, checkboxes, text areas, and submit buttons.

Q13 (15M) **Explain JSP implicit objects and session tracking in Web Applications.**

Implicit Objects: `request`, `response`, `session`, `application`, `out`, `config`, `pageContext`, `page`, `exception`. Session tracking via Cookies, URL Rewriting, and HttpSession.
